Bio173 Plant data worksheet- 15 points, due Friday 4/5

Start by opening the file "Plant Data W24.xlsx" on Canvas. The file should open to the "all sect" tab. This is a cut and paste from individual section data, captured in other tabs. Where pots were missing or not recorded in an individual section, "na" was entered into the sheet.

Each row represents data from one section, grouped into sections by plant. The top half represents sections where no fertilizer was added during growth; the bottom half shows data from sections where fertilizer was applied during growth. At the very bottom are statistics comparing growth with and without fertilizer added.

1) Looking at the number of plants measured for each pot, there clearly are some irregularities (yellow shading).

    a. Why might the plants measured be significantly lower than what was planted?

    b. Why might the plants measured be higher than what was intended?

    c. How would this affect the results?

2) Looking at the length and weight data, in most cases data for one type of pot were similar between sections. However, here too there are outliers (orange shading).

    a. What might cause these errors?

    b. Would it be OK to delete the yellow or orange data from the data set?

3) Compare numbers without fertilizer added to those with fertilizer.

    a. Is there a significant difference ($p < 0.05$) in number of plants germinated? Does that make sense?

    b. Now look at the statistics for length and weight. Does added fertilizer help plants grow? Speculate what might be going on.

4) Go to the tab for oats & tomatoes. This worksheet uses the summary numbers for those two species.
    a. Looking at the numbers for mean length and mean weight, what pattern do you see as the pots shift from all oat to all tomato? Which species is the stronger competitor? Does added fertilizer change this?

    b. What do the deWit series ("total yield") graphs tell you?

5) What is the competitive hierarchy between the three species? You can figure this out by looking at the pairwise comparisons.

6) Adding fertilizer could change the pattern of competition in one of two ways:

   i) by adding nutrients, you lessen the need to compete for them

   ii) by adding nutrients, you increase the advantage of the superior competitor

   From the data, which effect seems to be happening? Explain briefly.

7) What changes would you make to the protocol to reduce the incidence of error?

8) Does deleting erroneous data (yellow or orange cells) change the results at all? Try clearing the contents of these cells. The spreadsheet will recalculate numbers and redraw graphs.

9) What resources are the plants competing over? What does added fertilizer provide?

10) What lessons could a farmer draw from these results?